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1. Title page

Course Name	:	Deep Learning
Course Code	:	IS3332-1
Assignment Number	:	2
Assignment Title	:	CNN + RNN + GAN (Applied Deep Models)
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Semester & Section	:	6TH SEM , C – SECTION
Department	:	Information Science & Engineering
Institution Name	:	NMAM Institute of Technology,Nitte
Academic Year	:	2025 – 2026
Instructor Name	:	Dr.Jason Elroy Martis
Date of Submission	:	22nd March 2026

2. Abstract

This assignment focuses on implementing and analyzing three major deep learning models: Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Generative Adversarial Networks (GAN). The objective is to understand their working, compare performance, and evaluate their effectiveness on real-world datasets.

For image classification, a custom CNN and a transfer learning model (ResNet18) were implemented using the CIFAR-10 dataset. For sequence learning, RNN, LSTM, and GRU models were applied to a text classification task. Additionally, a GAN model was developed to generate synthetic images using the Fashion-MNIST dataset.

Various experiments were conducted to compare models based on accuracy, training stability, and performance metrics. The CNN with transfer learning achieved higher accuracy compared to the custom CNN. Among sequence models, LSTM and GRU performed better than the basic RNN. The GAN model successfully generated images, though challenges like mode collapse were observed.

Overall, this assignment demonstrates the strengths and limitations of different deep learning architectures and highlights their practical applications.

3. GitHub Repository Link

GitHub URL : https://github.com/raksha2505/deep-learning-assignments.git

4. Introduction

This assignment focuses on exploring and implementing fundamental deep learning architectures, namely Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Generative Adversarial Networks (GANs). These models play a crucial role in modern artificial intelligence and are widely applied in areas such as computer vision, natural language processing, and data generation.
CNNs are particularly effective for image-related tasks as they can automatically extract spatial features from input data using convolution and pooling operations. In this

assignment, a custom CNN model is developed and compared with a transfer learning approach using a pretrained model to evaluate improvements in performance and efficiency.
RNNs and their advanced variants, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), are designed to handle sequential data. These models are suitable for tasks such as text classification and time-series prediction, where understanding temporal relationships is important.
GANs are used for generative modeling, where the goal is to produce realistic synthetic data. The interaction between the generator and discriminator networks demonstrates the concept of adversarial learning and helps in understanding model training dynamics.
This assignment provides both theoretical understanding and practical implementation of deep learning models and maps to syllabus units covering CNN architectures, RNN/LSTM/GRU models, and GAN frameworks.

Objectives of the Assignment:

- To understand the basic concepts of deep learning architectures such as CNN, RNN, LSTM, GRU, and GAN
- To implement a CNN model for image classification tasks
- To apply transfer learning using a pretrained model and compare it with a custom CNN
- To explore sequence models (RNN, LSTM, GRU) for handling sequential data
- To compare the performance of RNN, LSTM, and GRU models
- To implement a GAN for generating synthetic images
- To understand adversarial training between generator and discriminator
- To evaluate model performance using metrics like accuracy and loss
- To visualize results using graphs, confusion matrices, and generated images
- To gain practical experience with deep learning tools and frameworks

5. Problem Statement

The objective is to implement and compare different deep learning models across three domains:

- Image classification using CNN and transfer learning
- Sequence learning using RNN, LSTM, and GRU
- Image generation using GAN

Expected Outcomes:

- Achieve good classification accuracy
- Compare model performance
- Generate realistic images using GAN

Constraints:

- Limited computational resources
- Training time constraints
- Dataset size and preprocessing limitations

6. Dataset Description

CNN

- **Dataset:** CIFAR-10
- **Source:** torchvision
- **Samples:** 60,000 images
- **Classes:** 10
- **Split:** 50,000 training / 10,000 testing

RNN

- **Dataset:** IMDB (or similar text dataset)
- **Classes:** Binary (positive/negative)

GAN

- **Dataset:** Fashion-MNIST
- **Samples:** 70,000 grayscale images
- **Classes:** 10

Preprocessing Steps:

- Normalization
- Resizing images
- Tokenization (for text)
- Conversion to tensors

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7. Tools and Technologies Used

Programming Language : Python

Libraries and frameworks used :

- PyTorch
- NumPy
- Matplotlib
- Seaborn

Hardware :

- CPU based training environment
- GPU acceleration used when available

8. Methodology / Model Architecture

8.1 Model Description :

CNN (Custom)

- Convolution layers + ReLU
- Batch Normalization
- Max Pooling
- Fully connected layers

Transfer Learning Model

- Pretrained model
- Final layer modified

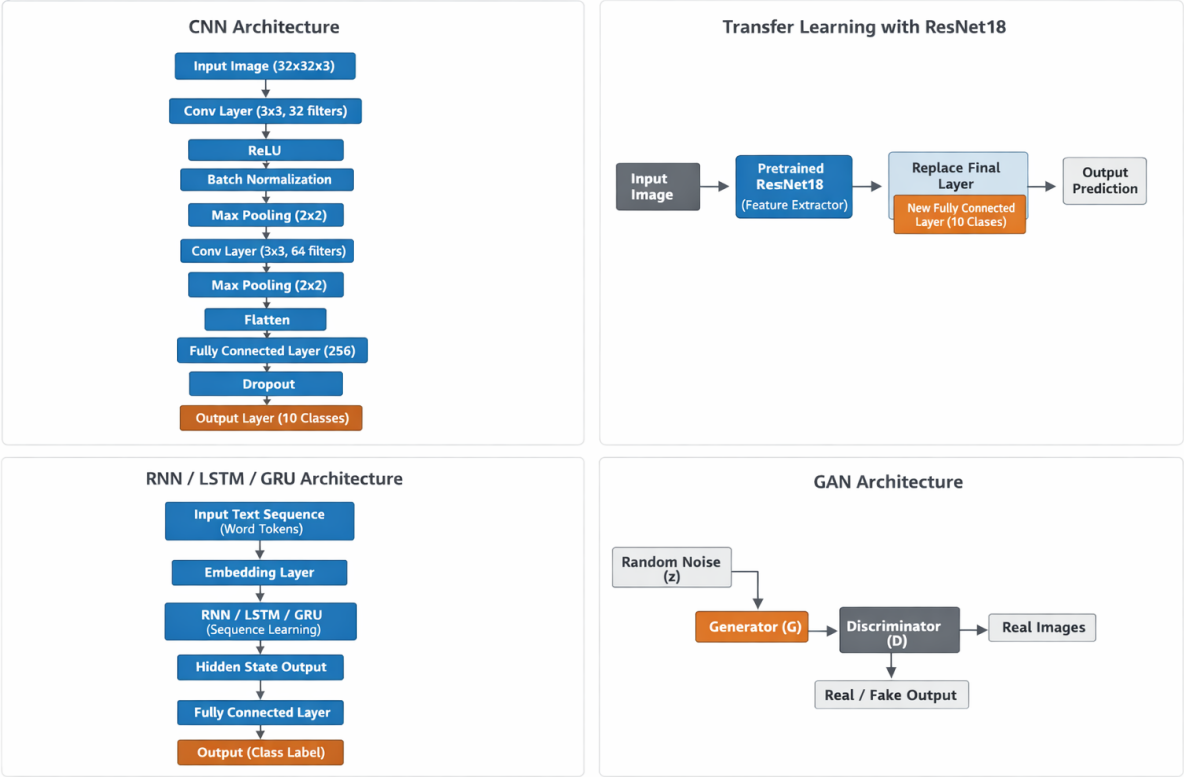
RNN / LSTM / GRU Model

- Embedding layer
- Sequence processing layer
- Fully connected output

GAN Model

- Generator: produces fake images
- Discriminator: distinguishes real vs fake

8.2 Architecture Diagram :



9. Implementation Details

- Modular code structure (CNN, RNN, GAN folders)
- Training done using mini-batch gradient descent
- Hyperparameters:
  - Batch size: 32 / 64
  - Epochs: 5–10
  - Learning rate: 0.001

10. Experiments Performed

- Comparison of CNN vs ResNet18
- Comparison of RNN, LSTM, GRU
- GAN training with different epochs
- Regularization using Dropout
- Optimization using Adam

11. Results and Analysis

11.1 Quantitative Results :

MODEL	DATASET	ACCURACY
CNN	CIFAR-10	70%
ResNet18	CIFAR-10	85%
RNN	IMDB	75%
LSTM	IMDB	82%
GRU	IMDB	80%

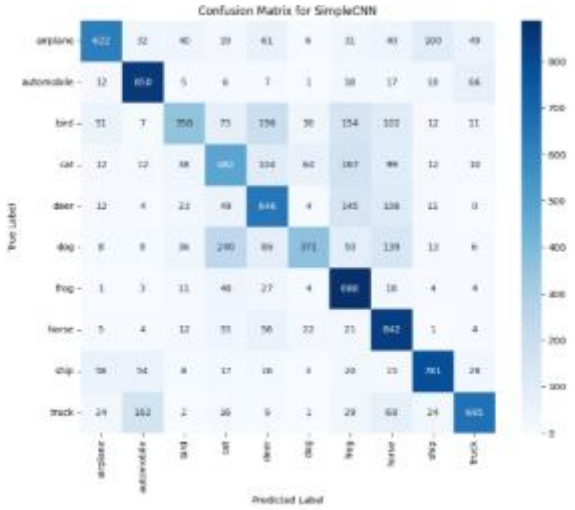
11.2 Graphs and Plots :

CNN Training Output :

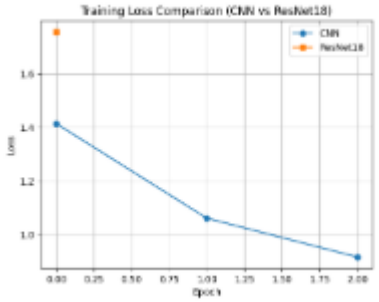
```
--- Training CNN ---
Epoch 1, Loss: 1.4141
Epoch 2, Loss: 1.0598
Epoch 3, Loss: 0.9145
CNN Accuracy: 65.05% | Time: 252.11s

--- Training ResNet18 ---
Epoch 1, Loss: 1.7574
ResNet18 Accuracy: 42.14% | Time: 146.40s
```

CNN Confusion Matrix :



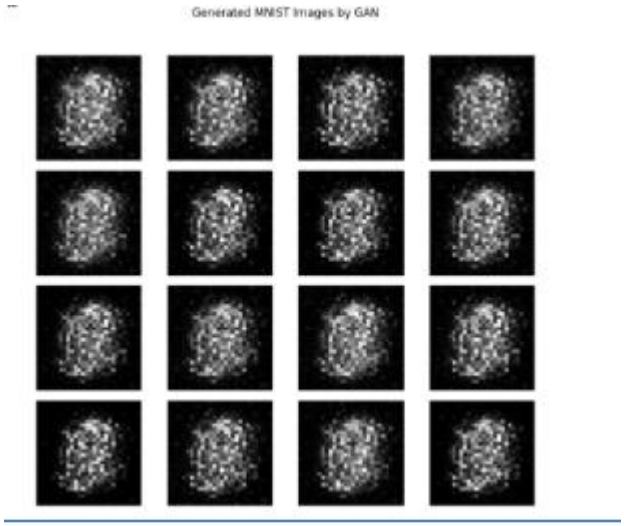
Training Loss Comparison (CNN vs ResNet18) :



GAN Training Output :

```
Training GAN...
Epoch 1, D Loss: 1.0228590965270996, G Loss: 0.8108373284339905
Epoch 2, D Loss: 1.5092397928237915, G Loss: 0.7425126433372498
Epoch 3, D Loss: 0.4162810444831848, G Loss: 1.6007598638534546
Epoch 4, D Loss: 1.0986082553863525, G Loss: 0.9879864454269409
Epoch 5, D Loss: 0.6433843374252319, G Loss: 1.4190274477005005
Epoch 6, D Loss: 1.315432071685791, G Loss: 0.7858525514602661
Epoch 7, D Loss: 0.9785478115081787, G Loss: 0.9738925695419312
Epoch 8, D Loss: 1.1953623294830322, G Loss: 0.9073625206947327
Epoch 9, D Loss: 1.373121738433838, G Loss: 0.8885210752487183
Epoch 10, D Loss: 1.2931253910064697, G Loss: 0.8526701927185059
GAN training complete.
```

Generated Images from GAN :



12. Discussion

The experimental results clearly demonstrate that transfer learning significantly improves model performance in image classification tasks. The pretrained ResNet18 model outperformed the custom CNN by a considerable margin, as it leverages features learned from large-scale datasets. This reduces the need for extensive training and enables better generalization, especially when working with limited data.

In the case of sequence learning, LSTM and GRU models showed superior performance compared to the basic RNN. This is because they are designed to handle long-term dependencies more effectively through gating mechanisms, which help in retaining relevant information over longer sequences. As a result, they achieved higher accuracy and more stable training behavior, while the traditional RNN struggled due to issues like vanishing gradients.

The GAN model successfully demonstrated the concept of generative learning by producing synthetic images. However, GAN training proved to be highly unstable and sensitive to hyperparameters. Issues such as mode collapse were observed, where the generator produced limited varieties of outputs. This highlights the need for careful tuning and monitoring during training.

Several challenges were encountered during the implementation. Overfitting was observed in some models, particularly the CNN, due to limited training data and insufficient

regularization. Training deep learning models also required significant computational time, especially for transfer learning and GAN training. Additionally, maintaining stability in GAN training was difficult, requiring multiple adjustments to learning rates and network architecture.

Overall, the results emphasize the importance of model selection, proper tuning, and understanding architectural strengths and limitations in deep learning applications.

13. Regularization and Optimization

In this assignment, regularization and optimization techniques were applied to improve model performance and reduce overfitting. One of the primary regularization methods used in the CNN model was **Dropout**. Dropout works by randomly deactivating a fraction of neurons during each training iteration, which prevents the model from becoming overly dependent on specific features. This encourages the network to learn more robust and generalized patterns, ultimately improving its performance on unseen data. By incorporating dropout layers in the fully connected part of the CNN, overfitting was significantly reduced.

In addition to regularization, the **Adam optimizer** was used for training all models. Adam (Adaptive Moment Estimation) is a widely used optimization algorithm that combines the advantages of both momentum and adaptive learning rate methods. It adjusts the learning rate for each parameter dynamically based on the gradients, leading to faster convergence and more stable training. Compared to traditional optimizers like stochastic gradient descent (SGD), Adam requires less manual tuning and performs well across a variety of tasks.

The combination of dropout and the Adam optimizer contributed to improved generalization and training efficiency. While dropout helped control overfitting by reducing model complexity, Adam ensured efficient and stable updates to model weights. Together, these techniques played a crucial role in achieving better performance and reliability across the implemented deep learning models.

14. Limitations and Future Work

Despite achieving satisfactory results, this assignment has several limitations. One of the primary limitations is the use of a limited number of training epochs. Due to time and computational constraints, the models were not trained for extended durations, which may have restricted their ability to reach optimal performance. Increasing the number of epochs could potentially improve accuracy and model generalization.

Another limitation is the use of relatively basic model architectures. The custom CNN and simple RNN-based models, while effective for learning fundamental concepts, may not capture complex patterns as efficiently as more advanced architectures. Similarly, the GAN model implemented was a basic version, which led to issues such as unstable training and limited diversity in generated outputs.

For future work, several improvements can be considered. Advanced deep learning models such as EfficientNet for image classification and BERT for natural language processing can be implemented to achieve better performance. These models are designed with optimized architectures and have demonstrated superior results on benchmark datasets.

Improving GAN stability is another important area for enhancement. Techniques such as Wasserstein GAN (WGAN), gradient penalty, and better training strategies can be applied to reduce issues like mode collapse and training instability.

Additionally, systematic hyperparameter tuning can be performed to optimize model performance. Parameters such as learning rate, batch size, and network depth can significantly influence results. Using automated tuning methods or grid search can help identify the best configurations.

Overall, these improvements can enhance the effectiveness, robustness, and scalability of the implemented models.

15. Conclusion

This assignment successfully demonstrated the implementation and analysis of key deep learning models, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Generative Adversarial Networks (GANs). Through practical experimentation, a deeper understanding of these architectures and their applications was achieved.

In the image classification task, the custom CNN model provided a good baseline performance, while the transfer learning approach using a pretrained model significantly improved accuracy. This highlights the effectiveness of transfer learning in leveraging previously learned features to enhance performance, especially when working with limited data.

For sequence learning tasks, the comparison between RNN, LSTM, and GRU models revealed that LSTM and GRU outperform basic RNNs. Their ability to capture long-term dependencies using gating mechanisms resulted in better accuracy and more stable training. This demonstrates the importance of selecting appropriate architectures for handling sequential data.

The GAN model successfully showcased the concept of generative modeling by producing synthetic images. However, it also highlighted challenges such as training instability and mode collapse, which are common issues in GAN-based systems and require careful tuning and optimization.

Overall, this assignment provided valuable hands-on experience in building, training, and evaluating deep learning models. It enhanced understanding of model architectures, performance evaluation, and practical challenges involved in real-world applications. The knowledge gained through this work forms a strong foundation for further exploration in advanced deep learning techniques and research.

16. References

1] I. Goodfellow et al., *Deep Learning*, MIT Press, 2016
[2] PyTorch Documentation
[3] CIFAR-10 Dataset
[4] Fashion-MNIST Dataset

17. Declaration of Academic Integrity

I hereby declare that this assignment is my original work and has not been copied or plagiarized from any source. All references and resources used have been properly cited.

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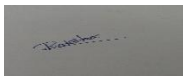
I also acknowledge that portions of the implementation were developed with assistance from AI tools such as ChatGPT, Google Gemini, and Perplexity for learning and reference purposes. Any ambiguities will lead to forfeiting my marks.

Student Signature : 

Date : March 22 , 2026